

```
PS C:\Users\GONZALO APAZA\Desktop\CLASES U\Catolica\7mo Semestre\Tecnologias Web 2\1erPrcial\Northwind> npm run db:check
```

```
> northwind-orders-api@1.0.0 db:check
> node scripts/check-db.js
```

Verificacion de la base "northwind" (entorno development)

1) Conexion

```
OK base "northwind" como usuario "postgres"
OK PostgreSQL 17.6 on x86_64-windows
```

2) Tablas requeridas en el esquema public

```
OK public.customers
OK public.employees
OK public.products
OK public.orders
OK public.order_details
OK public.shippers
```

3) Registros existentes (informativo)

```
OK customers: 91
OK employees: 9
OK products: 80
OK orders: 832
OK order_details: 2157
OK shippers: 6
```

4) Generacion de orders.order_id

```
OK tipo: smallint
OK DEFAULT: nextval('orders_order_id_seq'::regclass)
OK secuencia orders_order_id_seq (smallint), last_value=11080
OK MAX(order_id) actual = 11080
OK capacidad restante: 21687 identificadores
```

5) Trigger de inventario

```
OK trg_descontar_stock habilitado (tgenabled=0) -> descontar_stock()
```

Resultado: la base esta lista para la API.

```
PS C:\Users\GONZALO APAZA\Desktop\CLASES U\Catolica\7mo Semestre\Tecnologias Web 2\1erPrcial\Northwind> npm run
```

Lifecycle scripts included in northwind-orders-api@1.0.0:

```
start
  node src/server.js
test
  cross-env NODE_ENV=test node --test --test-concurrency=1 "tests/**/*.test.js"
```

available via `npm run`:

```
dev
  node --watch src/server.js
db:check
  node scripts/check-db.js
db:migrate
  node scripts/migrate.js
test:db:setup
  cross-env NODE_ENV=test node scripts/setup-test-db.js
test:one
  cross-env NODE_ENV=test node --test --test-concurrency=1
```

```
PS C:\Users\GONZALO APAZA\Desktop\CLASES U\Catolica\7mo Semestre\Tecnologias Web 2\1erPrcial\Northwind> []
```

```

Resultado: la base esta lista para la API.
PS C:\Users\GONZALO APAZA\Desktop\CLASES U\Catolica\7mo Semestre\Tecnologias Web 2\1erPrcial\Northwind> npm run
Lifecycle scripts included in northwind-orders-api@1.0.0:
  start
    node src/server.js
  test
    cross-env NODE_ENV=test node --test --test-concurrency=1 "tests/**/*.test.js"
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    node --watch src/server.js
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    node scripts/migrate.js
  test:db:setup
    cross-env NODE_ENV=test node scripts/setup-test-db.js
  test:one
    cross-env NODE_ENV=test node --test --test-concurrency=1
PS C:\Users\GONZALO APAZA\Desktop\CLASES U\Catolica\7mo Semestre\Tecnologias Web 2\1erPrcial\Northwind> npm start

> northwind-orders-api@1.0.0 start
> node src/server.js

[startup] conectado a la base "northwind"
[startup] PostgreSQL 17.6 on x86_64-windows
[startup] API escuchando en http://localhost:3000 (development)

```

localhost:3000

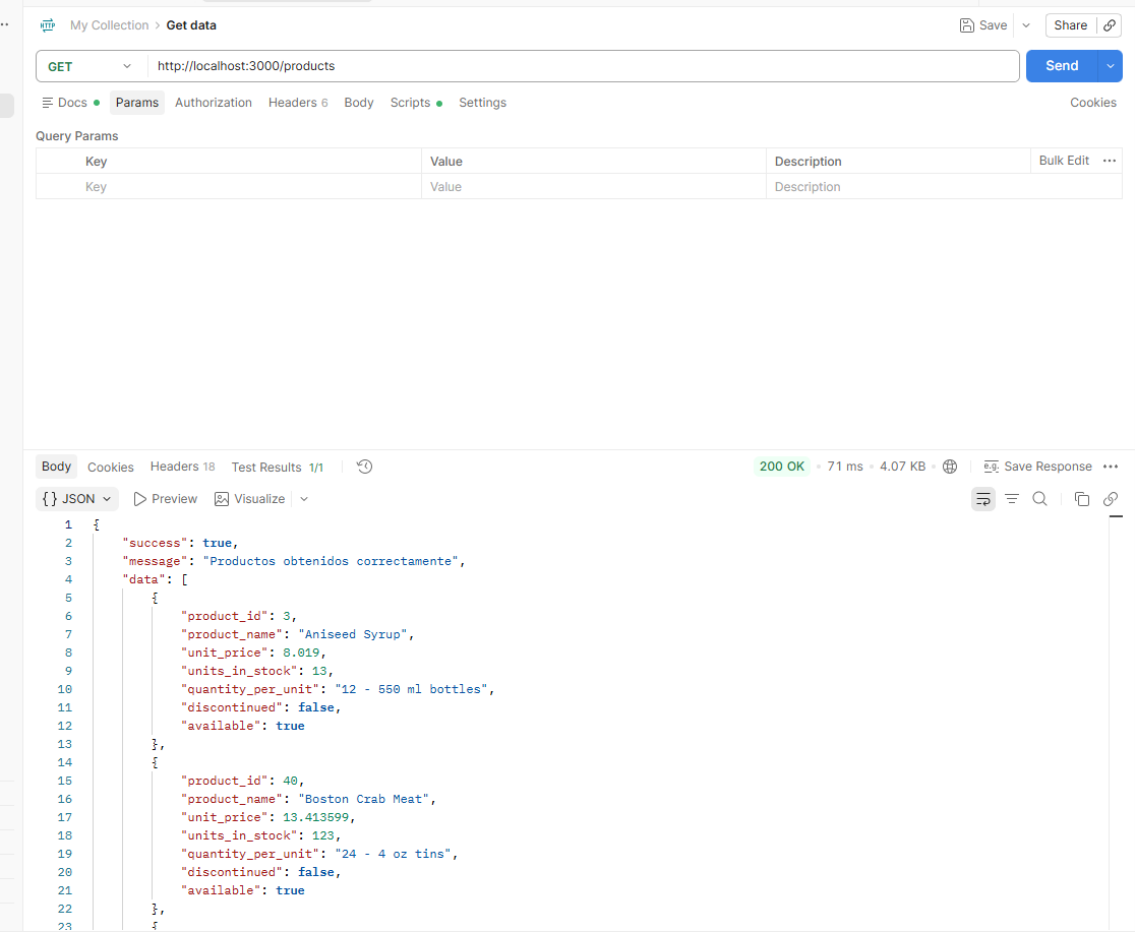
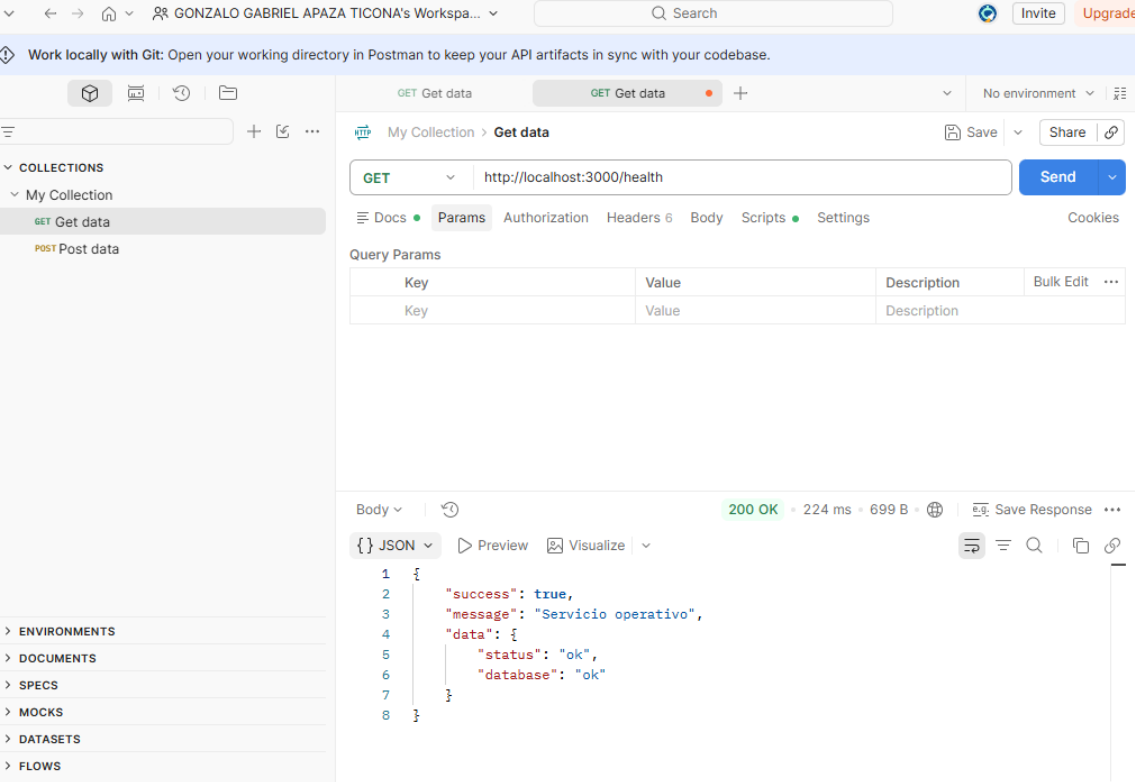
an formato al texto

```

"success":true,"message":"API de ordenes Northwind","data":{"endpoints":["GET /health","GET /customers","GET /employees","GET /products","GET /shippers","POST /orders","GET /orders/:id"]}}

```

Postman



```
Body Cookies Headers 18 Test Results 1/1 200 OK - 71 ms - 4.07 KB Save Response ...
{} JSON Preview Visualize
23 {
24   "product_id": 60,
25   "product_name": "Camembert Pierrot",
26   "unit_price": 24.786001,
27   "units_in_stock": 19,
28   "quantity_per_unit": "15 - 300 g rounds",
29   "discontinued": false,
30   "available": true
31 },
32 {
33   "product_id": 18,
34   "product_name": "Carnarvon Tigers",
35   "unit_price": 45.5625,
36   "units_in_stock": 42,
37   "quantity_per_unit": "16 kg pkg.",
38   "discontinued": false,
39   "available": true
40 },
41 {
42   "product_id": 39,
43   "product_name": "Chartreuse verte",
44   "unit_price": 14.4342,
45   "units_in_stock": 69,
```

```
Body Cookies Headers 18 Test Results 1/1 200 OK - 71 ms - 4.07 KB Save Response ...
{} JSON Preview Visualize
46   "units_in_stock": 07,
47   "quantity_per_unit": "750 cc per bottle",
48   "discontinued": false,
49   "available": true
50 },
51 {
52   "product_id": 4,
53   "product_name": "Chef Anton's Cajun Seasoning",
54   "unit_price": 17.641802,
55   "units_in_stock": 53,
56   "quantity_per_unit": "48 - 6 oz jars",
57   "discontinued": false,
58   "available": true
59 },
60 {
61   "product_id": 48,
62   "product_name": "Chocolade",
63   "unit_price": 9.29475,
64   "units_in_stock": 15,
65   "quantity_per_unit": "10 pkgs.",
66   "discontinued": false,
67   "available": true
68 },
```

My Collection - Get data Save Share

GET http://localhost:3000/products Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Key	Value	Description	Bulk Edit
Key	Value	Description	

```
Body Cookies Headers 18 Test Results 1/1 200 OK - 71 ms - 4.07 KB Save Response ...
{} JSON Preview Visualize
129   "available": true
130 },
131 {
132   "product_id": 6,
133   "product_name": "Grandma's Boysenberry Spread",
134   "unit_price": 20.0475,
135   "units_in_stock": 120,
136   "quantity_per_unit": "12 - 8 oz jars",
137   "discontinued": false,
138   "available": true
139 },
140 {
141   "product_id": 37,
142   "product_name": "Gravad lax",
143   "unit_price": 18.953999,
144   "units_in_stock": 11,
145   "quantity_per_unit": "12 - 800 g pkgs.",
146   "discontinued": false,
147   "available": true
148 },
149 {
150   "product_id": 69,
```

GET Get data

GET Get data

No environment

My Collection > Get data

GEThttp://localhost:3000/customersSend

DocsParamsAuthorizationHeaders6BodyScriptsSettingsCookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

BodyCookiesHeaders18Test Results1/1

200 OK154 ms3.92 KB

Save Response

JSON

PreviewVisualize

```
1 {
2   "success": true,
3   "message": "Clientes obtenidos correctamente",
4   "data": [
5     {
6       "customer_id": "ALFKI",
7       "company_name": "Alfreds Futterkiste",
8       "contact_name": "Maria Anders",
9       "contact_title": "Sales Representative",
10      "city": "Berlin",
11      "country": "Germany"
12    },
13    {
14      "customer_id": "ANATR",
15      "company_name": "Ana Trujillo Emparedados y helados",
16      "contact_name": "Ana Trujillo",
17      "contact_title": "Owner",
18      "city": "M\u00e9xico D.F.",
19      "country": "Mexico"
20    },
21    {
22      "customer_id": "ANTON",
23      "company_name": "Antonin Moreno Taqueria".
```

My Collection > Get data

GEThttp://localhost:3000/customersSend

DocsParamsAuthorizationHeaders6BodyScriptsSettingsCookies

Query Params

BodyCookiesHeaders18Test Results1/1

200 OK154 ms3.92 KB

Save Response

JSON

PreviewVisualize

```
150   "customer_id": "DUMUN",
151   "company_name": "Du monde entier",
152   "contact_name": "Janine Labrune",
153   "contact_title": "Owner",
154   "city": "Nantes",
155   "country": "France"
156 },
157 {
158   "customer_id": "EASTC",
159   "company_name": "Eastern Connection",
160   "contact_name": "Ann Devon",
161   "contact_title": "Sales Agent",
162   "city": "London",
163   "country": "UK"
164 }
165 ],
166 "meta": {
167   "total": 91,
168   "limit": 20,
169   "offset": 0,
170   "returned": 20
171 }
172 }
```

Standing by for launch!

GET Get data

GET Get data

No environment

My Collection > Get data

GET

http://localhost:3000/employees

Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers 18 Test Results 1/1

200 OK • 157 ms • 1.27 KB

Save Response

JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Empleados obtenidos correctamente",
4   "data": [
5     {
6       "employee_id": 5,
7       "first_name": "Steven",
8       "last_name": "Buchanan"
9     },
10    {
11      "employee_id": 8,
12      "first_name": "Laura",
13      "last_name": "Callahan"
14    },
15    {
16      "employee_id": 1,
17      "first_name": "Nancy",
18      "last_name": "Davolio"
19    },
20    {
21      "employee_id": 9,
22      "first_name": "Anne",
23      "last_name": "Dodsworth"
24    }
25  ]
26 }
```

Standing by for launch!

My Collection > Get data

GET

http://localhost:3000/employees

Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers 18 Test Results 1/1

200 OK • 157 ms • 1.27 KB

Save Response

JSON Preview Visualize

```
35 {
36   "employee_id": 3,
37   "first_name": "Janet",
38   "last_name": "Leverling"
39 },
40 {
41   "employee_id": 4,
42   "first_name": "Margaret",
43   "last_name": "Peacock"
44 },
45 {
46   "employee_id": 6,
47   "first_name": "Michael",
48   "last_name": "Suyama"
49 }
50 ],
51 "meta": {
52   "total": 9,
53   "limit": 20,
54   "offset": 0,
55   "returned": 9
56 }
57 }
```

Standing by for launch!

- Post

http://localhost:3000/orders

POST http://localhost:3000/orders

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "customer_id": "ALFKI",
3   "employee_id": 5,
4   "order_date": "2026-09-09",
5   "items": [
6     {
7       "product_id": 3,
8       "quantity": 1,
9       "discount": 0
10    },
11    {
12      "product_id": 4,
13      "quantity": 1,
14      "discount": 0.1
15    }
16  ]
17 }
```

Response History

http://localhost:3000/orders

POST http://localhost:3000/orders

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

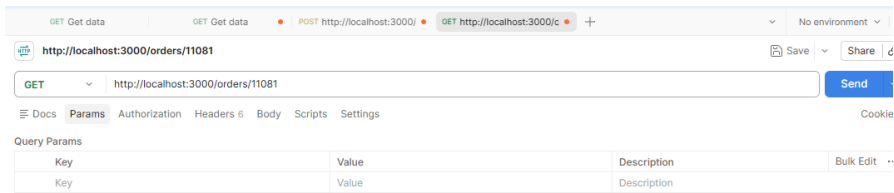
none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "customer_id": "ALFKI",
3   "employee_id": 5,
4   "order_date": "2026-09-09",
5   "items": [
6     {
7       "product_id": 3,
8       "quantity": 1,
9       "discount": 0
10    },
11    {
12      "product_id": 4,
13      "quantity": 1,
14      "discount": 0.1
15    }
16  ]
17 }
```

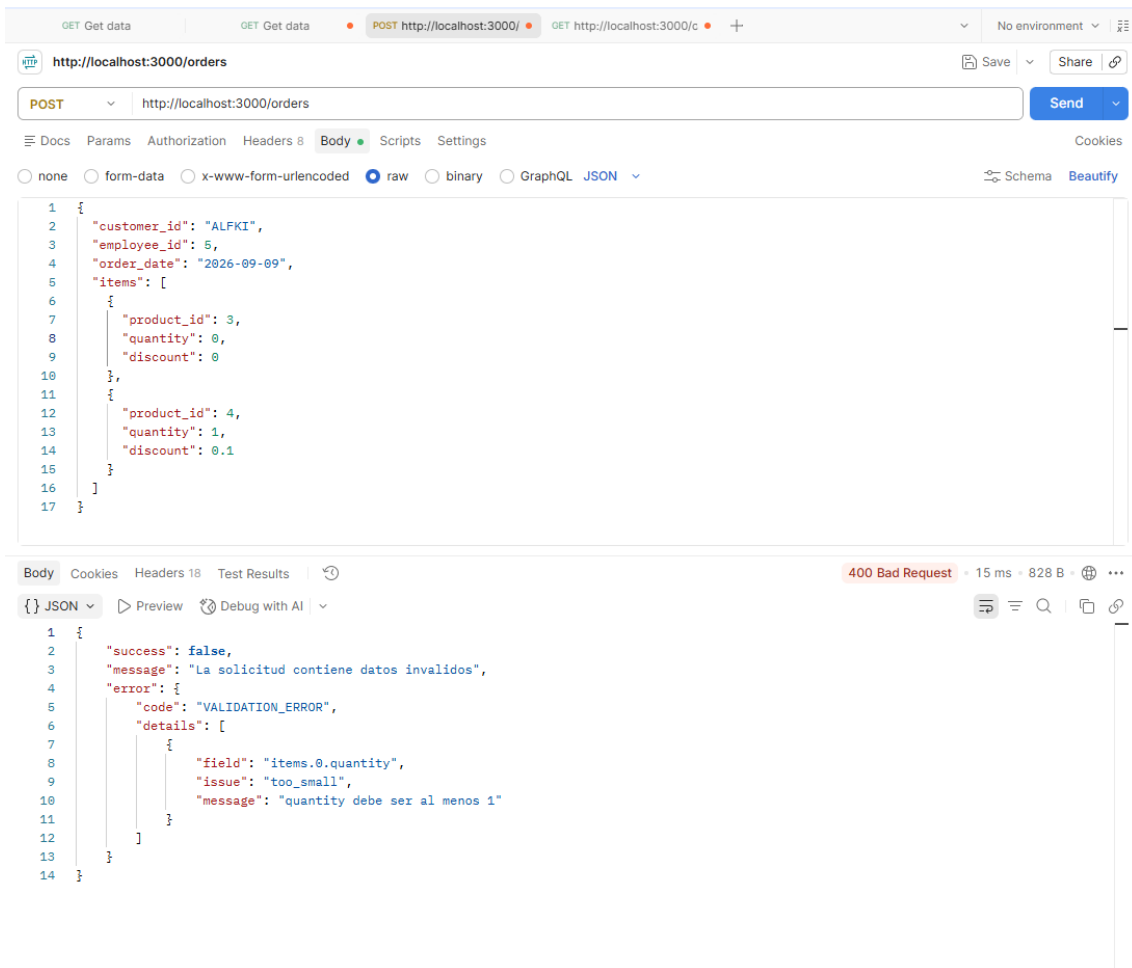
Body Cookies Headers 19 Test Results 201 Created · 569 ms · 1.55 KB

{ } JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Orden creada correctamente",
4   "data": {
5     "order_id": 11081,
6     "order_date": "2026-09-09",
7     "required_date": null,
8     "shipped_date": null,
9     "customer": {
10      "customer_id": "ALFKI",
11      "company_name": "Alfreds Futterkiste",
12      "contact_name": "Maria Anders"
13    },
14     "employee": {
15      "employee_id": 5,
16      "first_name": "Steven",
17      "last_name": "Buchanan"
18    },
19     "shipper": null,
20     "-----": "-----"
21   }
22 }
```



- Demostración de validación



Errores

GET Get dataGET Get dataPOST http://localhost:3000/GET http://localhost:3000/c

http://localhost:3000/orders

POSThttp://localhost:3000/ordersSend

DocsParamsAuthorizationHeadersBodyScriptsSettingsCookies

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSONSchemaBeautify

```
1 {
2   "customer_id": "ALFKI",
3   "employee_id": 5,
4   "order_date": "2026-09-09",
5   "items": [
6     {
7       "product_id": 9999,
8       "quantity": 1,
9       "discount": 0
10    }
11  ]
12 }
```

BodyCookiesHeaders18Test Results

404 Not Found172 ms750 B

{JSONPreviewDebug with AI

```
1 {
2   "success": false,
3   "message": "Uno o mas productos no existen",
4   "error": {
5     "code": "PRODUCT_NOT_FOUND",
6     "details": [
7       {
8         "product_id": 9999
9       }
10    ]
11  }
12 }
```

GET Get dataGET Get dataPOST http://localhost:3000/GET http://localhost:3000/c

http://localhost:3000/orders

POSThttp://localhost:3000/ordersSend

DocsParamsAuthorizationHeadersBodyScriptsSettingsCookies

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSONSchemaBeautify

```
1 {
2   "customer_id": "ALFKI",
3   "employee_id": 5,
4   "order_date": "2026-09-09",
5   "items": [
6     {
7       "product_id": 3,
8       "quantity": 30000,
9       "discount": 0
10    }
11  ]
12 }
```

BodyCookiesHeaders18Test Results

409 Conflict45 ms794 B

{JSONPreviewDebug with AI

```
1 {
2   "success": false,
3   "message": "Stock insuficiente para uno de los productos",
4   "error": {
5     "code": "INSUFFICIENT_STOCK",
6     "details": [
7       {
8         "product_id": 3,
9         "requested": 30000,
10        "available": 12
11      }
12    ]
13  }
14 }
```

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Servers (2)
 - PostgreSQL 17
 - Databases (10)
 - Portal_Propiedades
 - esapiens_lms_demo
 - hotel_oasis_resort
 - Inmobiliaria
 - kailith
 - kailith_test
 - northwind
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - northwind_test
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Query

```

1 SELECT *
2 FROM public.orders
3 ORDER BY order_id DESC
4 LIMIT 1;

```

Data Output

order_id [PK] smallint	customer_id character varying (5)	employee_id smallint	order_date date	required_date date	shipped_date date	ship_via smallint	freight real	ship_name character varying (40)	ship_address character varying
1	10311	ALFKI	5	2026-09-09	2026-09-16	[null]	2	5.5	Alfreds Futterkiste

Showing rows: 1 to 1 Page No: 1 of 1

pgAdmin 4

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 - Extensions
 - Foreign Data Wrappers
 - Languages

Query

```

1 SELECT *
2 FROM public.order_details
3 WHERE order_id = (
4     SELECT MAX(order_id)
5     FROM public.orders
6 )
7 ORDER BY product_id;

```

Data Output

order_id [PK] smallint	product_id [PK] smallint	unit_price real	quantity smallint	discount real
1	10311	1	18	0

Showing rows: 1 to 1 Page No: 1 of 1

pgAdmin 4

File Object Tools Edit View Window Help

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 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages

Query

```

3 p.product_name,
4 p.units_in_stock
5 FROM public.products p
6 WHERE p.product_id IN (
7     SELECT od.product_id
8     FROM public.order_details od
9     WHERE od.order_id = (
10         SELECT MAX(order_id)
11         FROM public.orders
12     )
13 )
14 ORDER BY p.product_id;

```

Data Output

product_id [PK] smallint	product_name character varying (40)	units_in_stock smallint
1	Chai	38

Showing rows: 1 to 1 Page No: 1 of 1

.- NPM TEST

```
Problems Output Debug Console Terminal Ports
PS C:\Users\GONZALO APAZA\Desktop\CLASES U\Catolica\7mo Semestre\Tecnologias Web 2\1erPrcial\Northwind> npm test

> northwind-orders-api@1.0.0 test
> cross-env NODE_ENV=test node --test --test-concurrency=1 "tests/**/*.test.js"

▶ GET /customers
  ✓ devuelve identificador, empresa y contacto con metadatos de paginacion (59.942ms)
  ✓ permite buscar por identificador (27.0256ms)
  ✓ permite buscar por nombre de empresa (9.9221ms)
  ✓ respeta limit y offset (19.7746ms)
  ✓ rechaza limit fuera de rango (9.7758ms)
  ✓ rechaza limit no numerico (11.9411ms)
  ✓ trata un intento de inyeccion SQL como texto de busqueda (20.5875ms)
✓ GET /customers (520.9978ms)
▶ GET /employees
  ✓ devuelve solo identificador, nombre y apellido (12.3072ms)
  ✓ no expone datos personales innecesarios (7.4781ms)
✓ GET /employees (20.4422ms)
▶ GET /products
  ✓ por defecto devuelve solo productos aptos para vender (18.4553ms)
  ✓ devuelve identificador, nombre, precio, stock y estado (10.2207ms)
  ✓ include_unavailable=true incluye descontinuados y sin stock (10.2917ms)
  ✓ rechaza include_unavailable con valor invalido (7.583ms)
✓ GET /products (47.1416ms)
▶ GET /shippers
  ✓ devuelve los transportistas disponibles (8.0327ms)
✓ GET /shippers (8.3054ms)
▶ Manejo centralizado de errores
  ✓ 400 e INVALID_JSON con un cuerpo JSON mal formado (49.0162ms)
  ✓ 415 cuando el Content-Type no es application/json (8.2848ms)
  ✓ 415 con formularios url-encoded (6.5427ms)
  ✓ 404 JSON (no HTML) para rutas inexistentes (5.3009ms)
  ✓ 404 JSON para metodos no implementados sobre una ruta existente (6.7446ms)
  ✓ las respuestas de error no filtran SQL, stack traces ni credenciales (7.7858ms)
  ✓ incluye cabeceras de seguridad de helmet y oculta x-powered-by (10.9056ms)
  ✓ GET /health confirma la conexion a la base (7.1048ms)
  ✓ rechaza un cuerpo JSON que excede el limite configurado (14.9292ms)
✓ Manejo centralizado de errores (217.7983ms)
▶ Politica de importes
  ✓ calcula bruto, descuento y total de linea (3.3683ms)
  ✓ evita los errores de la aritmetica binaria de JavaScript (0.4833ms)
  ✓ trata valores nulos como cero sin lanzar (0.2316ms)
  ✓ totaliza la orden y suma el flete (1.4039ms)
  ✓ flete NULL cuenta como 0 (0.2451ms)
  ✓ un descuento de 1 anula el total de la linea (0.2254ms)
✓ Politica de importes (7.6212ms)
▶ POST /orders - concurrencia
  ✓ dos solicitudes simultaneas que juntas superan el stock: solo una se confirma (391.0922ms)
  ✓ tres solicitudes simultaneas de 1 unidad sobre stock 2: exactamente 2 se confirman (179.818ms)
  ✓ solicitudes simultaneas reciben order_id distintos (193.1798ms)
✓ POST /orders - concurrencia (765.6315ms)
```

```
Problems  Output  Debug Console  Terminal  Ports

✓ POST /orders - concurrencia (765.6315ms)
▶ POST /orders - creacion exitosa
  ✓ crea cabecera y varios detalles, calcula importes y responde 201 con Location (129.0963ms)
  ✓ el order_id se genera en PostgreSQL y no colisiona con datos existentes (29.7846ms)
  ✓ ignora cualquier order_id, unit_price o total enviado por el cliente (15.9186ms)
  ✓ discount es opcional y por defecto 0 (27.8421ms)
  ✓ los campos opcionales pueden omitirse (29.5736ms)
✓ POST /orders - creacion exitosa (233.9064ms)
▶ POST /orders - validaciones de estructura
  ✓ rechaza cuerpo vacio (18.9157ms)
  ✓ rechaza un arreglo en lugar de un objeto (22.5046ms)
  ✓ rechaza campos desconocidos (19.2086ms)
  ✓ rechaza campos desconocidos dentro de items (17.5967ms)
  ✓ rechaza items vacio (23.1443ms)
  ✓ rechaza items que no son objetos (23.6649ms)
  ✓ rechaza customer_id vacio (20.38ms)
  ✓ rechaza customer_id de mas de 5 caracteres (24.7412ms)
  ✓ rechaza employee_id no entero, negativo o fuera del rango smallint (39.2787ms)
  ✓ rechaza cadenas numericas donde el contrato exige numeros (17.6518ms)
  ✓ rechaza fechas con formato invalido o inexistentes (37.6159ms)
  ✓ rechaza required_date anterior a order_date (15.7133ms)
  ✓ acepta required_date igual a order_date (35.4979ms)
  ✓ rechaza cantidad cero, negativa, decimal o fuera de rango (31.2252ms)
  ✓ rechaza descuento fuera del rango 0..1 y no finito (28.8002ms)
  ✓ acepta descuento en los limites 0 y 1 (38.5692ms)
  ✓ rechaza freight negativo (22.5743ms)
  ✓ rechaza campos de envio demasiado largos (24.7614ms)
  ✓ rechaza productos repetidos con mensaje claro (22.9454ms)
✓ POST /orders - validaciones de estructura (486.5766ms)
▶ POST /orders - entidades y disponibilidad
  ✓ 404 si el cliente no existe (28.8256ms)
  ✓ 404 si el empleado no existe (29.367ms)
  ✓ 404 si el transportista no existe (32.6472ms)
  ✓ 404 si el producto no existe (26.8702ms)
  ✓ 409 si el producto esta descontinuado (27.5093ms)
  ✓ 409 si el precio del producto es NULL (27.6414ms)
  ✓ 409 si el stock es NULL (NULL no es disponibilidad) (27.9952ms)
  ✓ 409 con detalle de disponibilidad si el stock es insuficiente (31.0229ms)
  ✓ un producto invalido cancela toda la orden (nada parcial) (31.3853ms)
  ✓ permite vender exactamente todo el stock disponible (38.8267ms)
✓ POST /orders - entidades y disponibilidad (303.3034ms)
▶ GET /orders/:id
  ✓ devuelve la orden creada con los mismos datos e importes (10.3565ms)
  ✓ incluye cabecera, cliente, empleado, transportista, detalles e importes (6.1323ms)
  ✓ usa los precios guardados en order_details, no los del catalogo actual (8.8132ms)
  ✓ devuelve 404 si la orden no existe (8.5242ms)
  ✓ devuelve 400 si el identificador tiene formato invalido (27.5618ms)
  ✓ lee tambien ordenes historicas cargadas directamente en la base (14.6569ms)
✓ GET /orders/:id (298.4805ms)
▶ POST /orders - rollback ante fallo real dentro de la transaccion
  ✓ un fallo posterior a los INSERT no deja registros parciales ni descuenta stock (108.1579ms)
```

```
Problems  Output  Debug Console  Terminal  Ports

✓ rechaza employee_id no entero, negativo o fuera del rango smallint (39.2787ms)
✓ rechaza cadenas numericas donde el contrato exige numeros (17.6518ms)
✓ rechaza fechas con formato invalido o inexistentes (37.6159ms)
✓ rechaza required_date anterior a order_date (15.7133ms)
✓ acepta required_date igual a order_date (35.4979ms)
✓ rechaza cantidad cero, negativa, decimal o fuera de rango (31.2252ms)
✓ rechaza descuento fuera del rango 0..1 y no finito (28.8002ms)
✓ acepta descuento en los límites 0 y 1 (38.5692ms)
✓ rechaza freight negativo (22.5743ms)
✓ rechaza campos de envio demasiado largos (24.7614ms)
✓ rechaza productos repetidos con mensaje claro (22.9454ms)
✓ POST /orders - validaciones de estructura (486.5766ms)
▶ POST /orders - entidades y disponibilidad
  ✓ 404 si el cliente no existe (28.8256ms)
  ✓ 404 si el empleado no existe (29.367ms)
  ✓ 404 si el transportista no existe (32.6472ms)
  ✓ 404 si el producto no existe (26.8702ms)
  ✓ 409 si el producto esta descontinuado (27.5093ms)
  ✓ 409 si el precio del producto es NULL (27.6414ms)
  ✓ 409 si el stock es NULL (NULL no es disponibilidad) (27.9952ms)
  ✓ 409 con detalle de disponibilidad si el stock es insuficiente (31.0229ms)
  ✓ un producto invalido cancela toda la orden (nada parcial) (31.3853ms)
  ✓ permite vender exactamente todo el stock disponible (38.8267ms)
✓ POST /orders - entidades y disponibilidad (303.3034ms)
▶ GET /orders/:id
  ✓ devuelve la orden creada con los mismos datos e importes (10.3565ms)
  ✓ incluye cabecera, cliente, empleado, transportista, detalles e importes (6.1323ms)
  ✓ usa los precios guardados en order_details, no los del catalogo actual (8.8132ms)
  ✓ devuelve 404 si la orden no existe (8.5242ms)
  ✓ devuelve 400 si el identificador tiene formato invalido (27.5618ms)
  ✓ lee tambien ordenes historicas cargadas directamente en la base (14.6569ms)
✓ GET /orders/:id (298.4805ms)
▶ POST /orders - rollback ante fallo real dentro de la transaccion
  ✓ un fallo posterior a los INSERT no deja registros parciales ni descuenta stock (108.1579ms)
  ✓ la conexion del pool queda utilizable despues del rollback (57.815ms)
✓ POST /orders - rollback ante fallo real dentro de la transaccion (236.8621ms)
i tests 74
i suites 12
i pass 74
i fail 0
i cancelled 0
i skipped 0
i todo 0
i duration_ms 7999.2932
PS C:\Users\GONZALO APAZA\Desktop\CLASES U\Catolica\7mo Semestre\Tecnologias Web 2\1erPrcial\Northwind> 
```